

Discrete Structures

Instructor: Sarfaraz Ahmed Baloch

Spring 2026

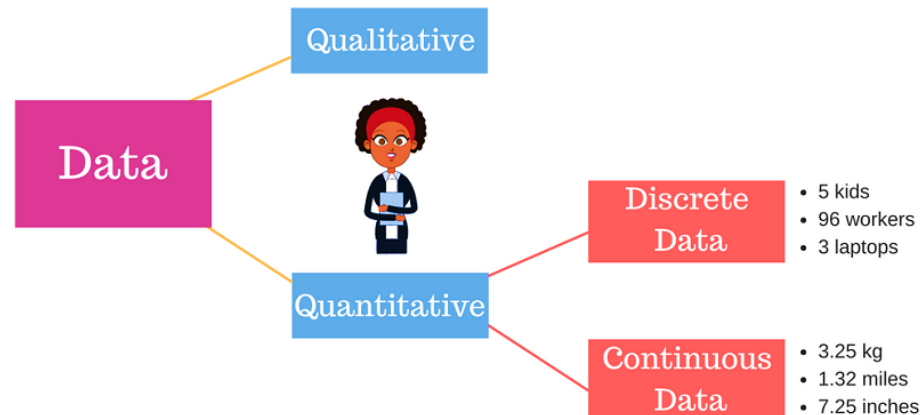
Aror University of Art, Architecture, Design & Heritage Sukkur

Discrete Structures vs Discrete Mathematics

Discrete Structures are structures that are used in describing discrete mathematics.

Discrete Mathematics is math that makes use of discrete structures.

"Discrete Math" OR "Discrete Structure" is not the name of a branch of mathematics, like number theory, algebra, calculus, etc. Rather, it's a description of a set of branches of math that all have in common the feature that they are "**discrete**" rather than "**continuous**".



Recommended Books

Text Book

- Discrete Mathematics and Its Applications

Kenneth H. Rosen

Including Solution Manual

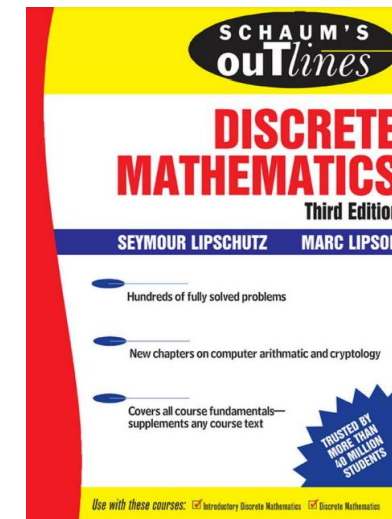
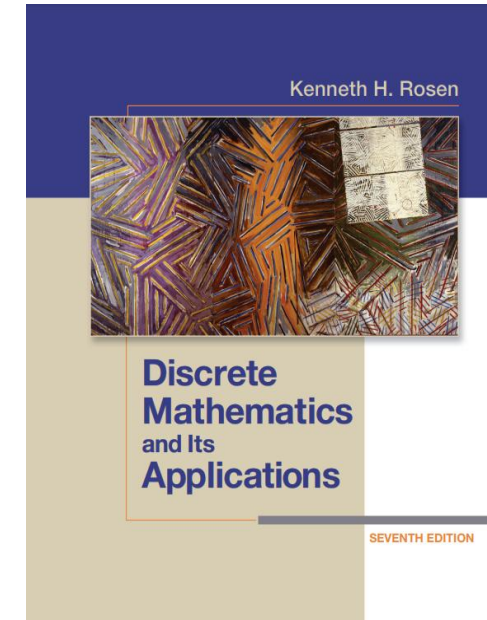
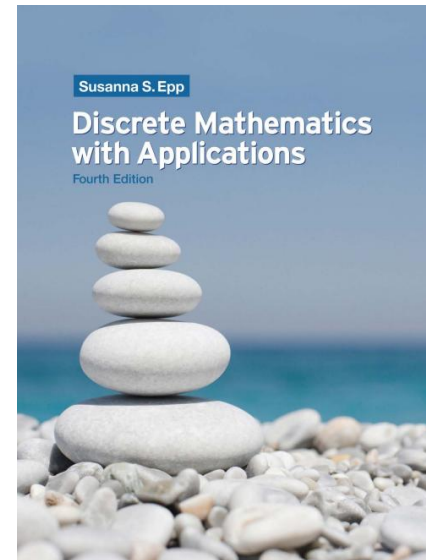
Reference Book:

- Discrete Mathematics with Applications

Susanna S.Epp

- Discrete Mathematics

Schaums' Outlines



Logic: Logic is the study of reasoning. It is the basic on which all the sentence are build. OR Correct and Incorrect reasoning.

A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Imperative

Declarative

Interrogative

1.What time is it?

2.Read this carefully.

3. $x + 1 = 2$.

4. $x + y = z$

5.I am in Class.

6.Cat is bigger than Rat.

7. $1 + 2 = 5$

8. $2 + 2 = 5$

Propositional Logic:

The area of logic that deals with propositions.

- “Elephants are bigger than mice.”

Is this a statement? **yes**

Is this a proposition? **yes**

What is the truth value
of the proposition? **true**

- “ $y > 5$ ”

Is this a statement? **yes**

Is this a proposition? **no**

Truth value?

Depends on the value of y , but this
value is not specified.

Compound Propositions

Logical Connectives

I like A.I

Logical operators are used to form new propositions also called compound propositions from two or more existing propositions.

- Negation $\neg$
- Conjunction $\wedge$
- Disjunction $\vee$
- Implication $\rightarrow$
- Biconditional $\leftrightarrow$

- Conjunction (AND)
- Disjunction (OR)
- Exclusive or (XOR)
- Implication (if – then)
- Biconditional (if and only if)

A **truth table** is a tabular representation of all the combinations of values for inputs and their corresponding outputs

Negation

- I am in NOT class
- Cat cannot fly.
- Today is Sunday.

The Truth Table for the Negation of a Proposition.	
p	$\neg p$
T	F
F	T

Conjunction

Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition “ p and q ”. The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Today is Friday AND it is raining today.
I'd like pizza AND a salad for lunch.

The Truth Table for the Conjunction of Two Propositions.		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition “ p or q ”. The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Inclusive or : The disjunction is true when at least one of the two propositions is true.

“Students who have taken calculus or computer science can take this class.”

Exclusive or : The disjunction is true only when one of the proposition is true.

“Ice cream or pudding will be served after lunch.”

The Truth Table for the Disjunction of Two Propositions.		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

The Truth Table for the Exclusive Or (XOR) of Two Propositions.		
p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

CONDITIONAL STATEMENT / IMPLICATION:

Let p and q be propositions. The conditional statement $p \rightarrow q$, is the proposition “if p , then q ” which is false when p is true and q is false, and true otherwise.

p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

Example: “If I am elected, then I will lower taxes.”

elected, lower taxes.

not elected, lower taxes.

not elected, not lower taxes.

elected, not lower taxes.

TABLE 5 The Truth Table for the Conditional Statement $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

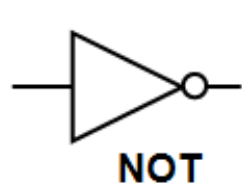
CONDITIONAL STATEMENT / Bi-implications:

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications

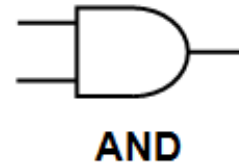
“You can take the flight if and only if you buy a ticket.”

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

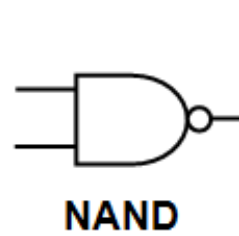
1.2.6 Logic Circuits – Logic Gate



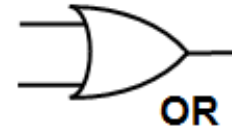
Input	Output
I	F
0	1
1	0



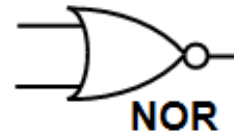
Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0



Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0

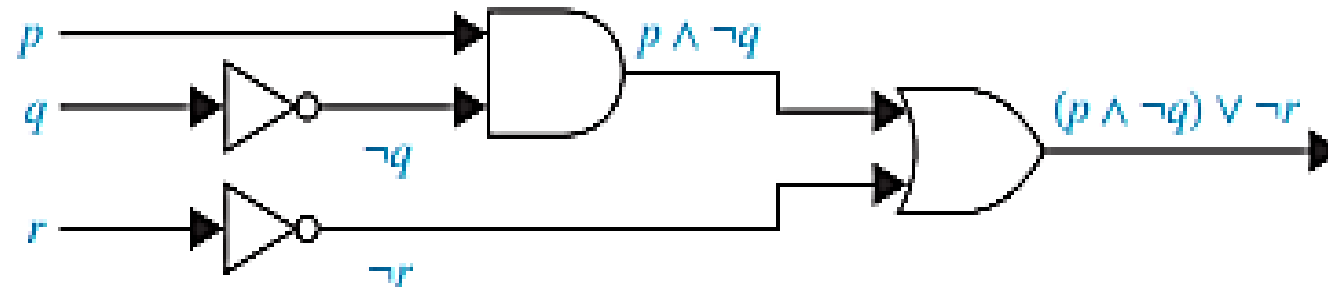


Inputs		Output
A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

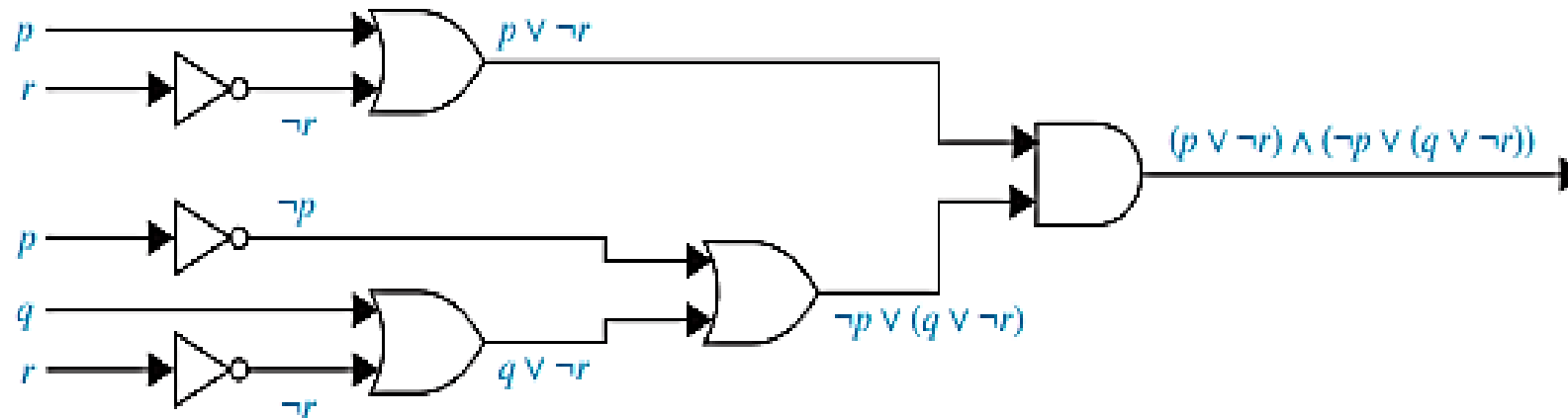


Inputs		Output
A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

1.2.6 Logic Circuits – Logic Gate



$$(p \wedge \neg q) \vee \neg r$$



$$(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r)).$$

Other Conditional Statements:

Conditional: “if p , then q ” (or $p \rightarrow q$)

- If you are working hard, then you are a topper.

Converse: “if q , then p ” (or $q \rightarrow p$)

- If you are a topper, then you are working hard.

Inverse: “if not p , then not q ” (or $\neg p \rightarrow \neg q$)

- If you are not working hard, then you are not a topper.

Contrapositive: “if not q , then not p ” (or $\neg q \rightarrow \neg p$)

- If you are not topper, then you are not working hard.

Show using a truth table that the conditional is equivalent to the contrapositive.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	$\textcircled{\text{T}}$	T

Show using truth tables that neither the **converse** nor **inverse** of an implication are not equivalent to the implication.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	$\textcircled{\text{T}}$	T	T

CONDITIONAL STATEMENT:

Different Ways of Expressing $p \rightarrow q$

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

“if p , then q ”

“if p , q ”

“ p is sufficient for q ”

“ q if p ”

“ q when p ”

“a necessary condition for p is q ”

“ q unless $\neg p$ ”

“ p implies q ”

“ p only if q ”

“a sufficient condition for q is p ”

“ q whenever p ”

“ q is necessary for p ”

“ q follows from p ”

Example: “If I am elected, then I will lower taxes.”

elected, lower taxes.

not elected, lower taxes.

not elected, not lower taxes.

elected, not lower taxes.

“p only if q”

This is often the most confusing one because “if” is right next to q. However, in logic, “only if” introduces the consequence (the result), not the condition.

The Logic: It means p cannot happen unless q is true. If you have p, you are guaranteed to have q.

Example: “You can drive only if you have a license.”

(a) If you are driving (p), then you must have a license (q).

(b) It does not mean if you have a license, you are currently driving.

“q unless not p”

The word “unless” can be translated as “if not”. So, “q unless not p” can be written as: “q if not (not p).”

The Logic: Two negatives cancel each other out ($\neg(\neg p) \equiv p$).

The Transformation:

1. Start with: q unless $\neg p$
2. Replace “unless” with “if not”: q if $\neg(\neg p)$
3. Simplify: q if p
4. Rearrange: If p, then q

Example: “The alarm will ring (q) unless the power is not on ($\neg p$).”

This means: If the power is on (p), the alarm will ring (q).

Practice Time

Write each of these statements in the form “if p, then q” in English.

[Hint: Refer to the list of common ways to express conditional statements provided in this section.]

- a) You send me an e-mail message only if I will remember to send you the address.
- b) To be a citizen of this country, it is sufficient that you were born in the Pakistan.
- c) If you keep your textbook, it will be a useful reference in your future courses.
- d) The Red Wings will win the Stanley Cup if their goalie plays well.
- e) That you get the job implies that you had the best credentials.
- f) The beach erodes whenever there is a storm.
- g) It is necessary to have a valid password to log on to the server.
- h) You will reach the summit unless you begin your climb too late.

Practice Time

Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express each of these propositions as an English sentence.

a) $p \rightarrow q$

b) $\neg q \leftrightarrow r$

c) $q \rightarrow \neg r$

d) $p \vee q \vee r$

e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$

f) $(p \wedge q) \vee (\neg q \wedge r)$

Practice Time

Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives.

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

Propositional Equivalences, Law of Logic

- A **tautology** is a proposition which is always true.

Example: $p \vee \neg p$

- A **contradiction** is a proposition which is always false.

Example: $p \wedge \neg p$

- A **contingency** is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Propositional Equivalences, Law of Logic

Show that the proposition form $p \vee \neg p$ is a tautology and the proposition form $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Exercise: If t is a tautology and c is contradiction, show that $p \vee t \equiv t$ and $p \wedge c \equiv c$?

Contradiction occurs when we get a statement p , such that p is true and its negation $\neg p$ is also true.

Truth Tables For Compound Propositions

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.					
p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Propositional Equivalences, Law of Logic

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

De Morgan's laws state that:

The negation of an **and** proposition is logically equivalent to the **or** proposition in which each component is negated.

$$1. \neg(p \wedge q) \equiv \neg p \vee \neg q$$

The negation of an **or** proposition is logically equivalent to the **and** proposition in which each component is negated.

$$2. \neg(p \vee q) \equiv \neg p \wedge \neg q$$

Question: Negate the following compound Propositions

- John is six feet tall **and** he weights at least 200 pounds.
- The bus was late **or** Tom's watch was slow.

Solution

- John is **not** six feet tall **or** he weighs **less** than 200 pounds.
- The bus was **not** late **and** Tom's watch was **not** slow.

Propositional Equivalences, Law of Logic

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$\begin{aligned}
 p \rightarrow q &\equiv \neg p \vee q \\
 p \rightarrow q &\equiv \neg q \rightarrow \neg p \\
 p \vee q &\equiv \neg p \rightarrow q \\
 p \wedge q &\equiv \neg(p \rightarrow \neg q) \\
 \neg(p \rightarrow q) &\equiv p \wedge \neg q \\
 (p \rightarrow q) \wedge (p \rightarrow r) &\equiv p \rightarrow (q \wedge r) \\
 (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (p \vee q) \rightarrow r \\
 (p \rightarrow q) \vee (p \rightarrow r) &\equiv p \rightarrow (q \vee r) \\
 (p \rightarrow r) \vee (q \rightarrow r) &\equiv (p \wedge q) \rightarrow r
 \end{aligned}$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$\begin{aligned}
 p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\
 p \leftrightarrow q &\equiv \neg p \leftrightarrow \neg q \\
 p \leftrightarrow q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \\
 \neg(p \leftrightarrow q) &\equiv p \leftrightarrow \neg q
 \end{aligned}$$

Propositional Equivalences, Law of Logic

Using laws of logic, show that

$$\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution

Take $\neg(\neg p \wedge q) \wedge (p \vee q)$

$$\begin{aligned} &\equiv (\neg(\neg p) \vee \neg q) \wedge (p \vee q) && \text{by Demorgan's Law} \\ &\equiv (p \vee \neg q) \wedge (p \vee q) && \text{by Double Negation Law} \\ &\equiv p \vee (\neg q \wedge q) && \text{by Distributive Law} \\ &\equiv p \vee F && \text{by Negation Law} \\ &\equiv p && \text{by Identity Law} \end{aligned}$$

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

Show $\neg(p \rightarrow q)$ is equivalent to $p \wedge \neg q$.

Show that $\neg(p \vee \neg(p \wedge q))$ is a contradiction

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$$

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

Prove that $\neg[r \vee (q \wedge (\neg r \rightarrow \neg p))] \equiv \neg r \wedge (p \vee \neg q)$

$$\begin{aligned}
 & \neg[r \vee (q \wedge (\neg r \rightarrow \neg p))] \\
 \equiv & \neg r \wedge \neg(q \wedge (\neg r \rightarrow \neg p)), && \text{De Morgan's law} \\
 \equiv & \neg r \wedge \neg(q \wedge (\neg r \vee \neg \neg p)), && \text{Conditional rewritten as disjunction} \\
 \equiv & \neg r \wedge \neg(q \wedge (r \vee \neg p)), && \text{Double negation law} \\
 \equiv & \neg r \wedge (\neg q \vee \neg(r \vee \neg p)), && \text{De Morgan's law} \\
 \equiv & \neg r \wedge (\neg q \vee (\neg r \wedge p)), && \text{De Morgan's law, double negation} \\
 \equiv & (\neg r \wedge \neg q) \vee (\neg r \wedge (\neg r \wedge p)), && \text{Distributive law} \\
 \equiv & (\neg r \wedge \neg q) \vee ((\neg r \wedge \neg r) \wedge p), && \text{Associative law} \\
 \equiv & (\neg r \wedge \neg q) \vee (\neg r \wedge p), && \text{Idempotent law} \\
 \equiv & \neg r \wedge (\neg q \vee p), && \text{Distributive law} \\
 \equiv & \neg r \wedge (p \vee \neg q), && \text{Commutative law}
 \end{aligned}$$

TABLE 6 Logical Equivalences.

Equivalence	Name
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

p	q	r	$p \vee \neg q$	$q \vee \neg r$	$r \vee \neg p$	$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$
T	T	T	T	T	T	T
T	T	F	T	T	F	F
T	F	T	T	F	T	F
T	F	F	T	T	F	F
F	T	T	F	T	T	F
F	T	F	F	T	T	F
F	F	T	T	F	T	F
F	F	F	T	T	T	T

Notation

$\bigvee_{j=1}^n p_j$ is used for $p_1 \vee p_2 \vee \dots \vee p_n$

$\bigwedge_{j=1}^n p_j$ is used for $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Predicates

Predicate Logic deals with predicates, which are propositions, consist of variables.

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables. The domain of a predicate variable is the set of all values that may be substituted in place of the variable.

Property of subject expression that contain variables

Why do we study it in DM, [to deal with variables]

Propositional Function

Predicate is also called propositional Function

Denote: $P(\text{Variable})$ or $P(x)$

- **Example 1:** Let $P(x)$ denote the statement " $x > 10$ ". What are the truth values of $P(11)$ and $P(5)$?

Solution: $P(11)$ is equivalent to the statement $11 > 10$, which is True.

$P(5)$ is equivalent to the statement $5 > 10$, which is False.

- **Example 2:** Let $R(x, y)$ denote the statement " $x = y + 1$ ". What is the truth value of the propositions $R(1, 3)$ and $R(2, 1)$?

Solution: $R(1, 3)$ is the statement $1 = 3 + 1$, which is False.

$R(2, 1)$ is the statement $2 = 1 + 1$, which is True.

Predicates

Consider the propositional function $Q(x, y, z)$ defined by:

$$x + y = z$$

What is the truth value of $Q(2, 3, 5)$?

True

What is the truth value of $Q(0, 1, 2)$?

False



$$2+3=5$$

$$5=5$$



$$0+1=2$$

$$1=2$$

Predicates

EXAMPLE 4 Let $A(c, n)$ denote the statement “Computer c is connected to network n ,” where c is a variable representing a computer and n is a variable representing a network. Suppose that the computer MATH1 is connected to network CAMPUS2, but not to network CAMPUS1. What are the values of $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$?

Solution: Because MATH1 is not connected to the CAMPUS1 network, we see that $A(\text{MATH1}, \text{CAMPUS1})$ is false. However, because MATH1 is connected to the CAMPUS2 network, we see that $A(\text{MATH1}, \text{CAMPUS2})$ is true. ◀

Similarly, we can let $R(x, y, z)$ denote the statement “ $x + y = z$.” When values are assigned to the variables x , y , and z , this statement has a truth value.

EXAMPLE 5 What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: The proposition $R(1, 2, 3)$ is obtained by setting $x = 1$, $y = 2$, and $z = 3$ in the statement $R(x, y, z)$. We see that $R(1, 2, 3)$ is the statement “ $1 + 2 = 3$,” which is true. Also note that $R(0, 0, 1)$, which is the statement “ $0 + 0 = 1$,” is false. ◀

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by

$$P(x_1, x_2, \dots, x_n).$$

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the **propositional function** P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an **n -place predicate** or a **n -ary predicate**.

Predicates and Quantifiers

The variable of predicates is quantified by quantifiers. There are two types of quantifier in predicate logic - **Existential Quantifier** and **Universal Quantifier**.

TABLE 1 Quantifiers.		
Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

- $\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

Quantifiers

- **Example 1:** Let $P(x)$ be the statement " $x + 2 > x$ ". What is the truth value of the statement $\forall x P(x)$?

Solution: As $x + 2$ is greater than x for any real number, so $P(x) \equiv T$ for all x or $\forall x P(x) \equiv T$.

-
- **Example :** Let $P(x)$ be the statement " $x > 5$ ". What is the truth value of the statement $\exists x P(x)$?

Solution: $P(x)$ is true for all real numbers greater than 5 and false for all real numbers less than 5. So $\exists x P(x) \equiv T$.

Quantifiers

- $\exists!x P(x)$ means that $P(x)$ is true for one and only one x in the universe of discourse.
- This is commonly expressed in English in the following equivalent ways:
 - “There is a unique x such that $P(x)$.”
 - “There is one and only one x such that $P(x)$ ”
- Examples:
 1. If $P(x)$ denotes “ $x + 1 = 0$ ” and U is the integers, then $\exists!x P(x)$ is true.
 2. But if $P(x)$ denotes “ $x > 0$,” then $\exists!x P(x)$ is false.
- The uniqueness quantifier is not really needed as the restriction that there is a unique x such that $P(x)$ can be expressed as:

$$\exists x (P(x) \wedge \forall y (P(y) \rightarrow y=x))$$

Quantifiers

Let $C(x)$ be the statement “ x has a cat,” let $D(x)$ be the statement “ x has a dog,” and let $F(x)$ be the statement “ x has a ferret.” Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

a) A student in your class has a cat, a dog, and a ferret.

Solution: $\exists x (C(x) \wedge D(x) \wedge F(x))$

b) All students in your class have a cat, a dog, and a ferret.

Solution: $\forall x (C(x) \wedge D(x) \wedge F(x))$

c) Some student in your class has a cat and a ferret, but not a dog.

Solution: $\exists x (C(x) \wedge F(x) \wedge \neg D(x))$

d) No student in your class has a cat, a dog, and a ferret.

Solution: $\forall x \neg(C(x) \wedge D(x) \wedge F(x))$

37. Express each of these statements using predicates and quantifiers.

- a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.
- b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.
- c) A student must take at least 60 course hours, or at least 45 course hours and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.
- d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Answer:

- a) $\forall x (f(x) \vee g(x) \rightarrow e(x))$
- b) $\forall x (m(x) \wedge c(x, 3.5)) \vee (\neg m(x) \wedge c(x, 3.5)) \rightarrow q(x)$
- c) $M \rightarrow ((H(60) \vee (H(45) \wedge W)) \wedge \forall y G(B, y))$
- d) $\exists x (C(x) \wedge A(x))$

a)

f(x) : x is flying more than 25000 miles in a year

g(x) : x takes more than 25 flights during that year

e(x) : x qualifies as an elite flyer

b)

m(x): x is a man

c(x, y) : x has the change if previous time is less than y

q(x) : qualifies for the marathon

c)

M: received Master's degree

H(x): took x hours for the course

W: wrote thesis

G(x, y) : graded x or higher in course y

d)

C(x) : x taken more than 21 credit hours in a semester

A(x) : x has received all A

Negating Quantified Expressions

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Quantifiers, De Morgan's Laws for Quantifiers

What are the negations of the statements “There is an honest politician” and “All Americans eat cheeseburgers”?

Solution: Let $H(x)$ denote “ x is honest.” Then the statement “There is an honest politician” is represented by $\exists x H(x)$, where the domain consists of all politicians. The negation of this statement is $\neg \exists x H(x)$, which is equivalent to $\forall x \neg H(x)$.

This negation can be expressed as “Every politician is dishonest.”

(Note: In English, the statement “All politicians are not honest” is ambiguous. In common usage, this statement often means “Not all politicians are honest.”

Consequently, we do not use this statement to express this negation.)

Quantifiers, De Morgan's Laws for Quantifiers

What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = x)$?

Solution:

The negation of $\forall x(x^2 > x)$ is the statement $\neg \forall x(x^2 > x)$, which is equivalent to $\exists x \neg(x^2 > x)$. This can be rewritten as $\exists x(x^2 \leq x)$.

The negation of $\exists x(x^2 = x)$ is the statement $\neg \exists x(x^2 = x)$, which is equivalent to $\forall x \neg(x^2 = x)$.

This can be rewritten as $\forall x(x^2 \neq x)$.

The truth values of these statements depend on the domain.

Show that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution:

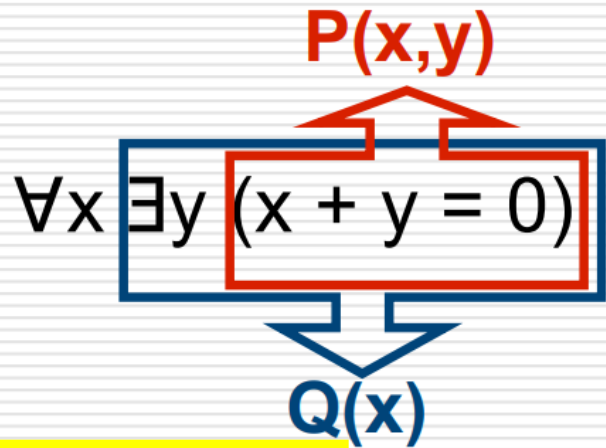
By De Morgan's law for universal quantifiers, we know that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(\neg(P(x) \rightarrow Q(x)))$ are logically equivalent.

By the fifth logical equivalence, we know that $\neg(P(x) \rightarrow Q(x))$ and $P(x) \wedge \neg Q(x)$ are logically equivalent for every x . Because we can substitute one logically equivalent expression for another in a logical equivalence, it follows that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Nested Quantifiers

Two quantifiers are nested if one is within the scope of the other.

$$\forall x \exists y (x + y = 0)$$



$$\forall x Q(x)$$

$$Q(x) \text{ is } \exists y P(x,y)$$

$$P(x,y) \text{ is } (x + y = 0)$$

$$\forall x \forall y (x + y = y + x)$$

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$$

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$$

Nested Quantifiers, Orders

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

$\forall x \forall y P(x, y)$

$\forall x \exists y P(x, y)$

$\exists x \forall y P(x, y)$

$\exists x \exists y P(x, y)$

So, we concluded the new concept of Equivalency here as well.

Nested Quantifiers, Questions

Translate these statements into English, where the domain for each variable consists of all real numbers.

a) $\exists x \forall y (xy = y)$

b) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$

c) $\forall x \forall y \exists z (x = y + z)$

a) This says that there exists a real number x such that for every real number y , the product xy equals y . That is, there is a multiplicative identity for the real numbers. This is a true statement, since $x = 1$ is the identity.

b) The product of two negative real numbers is always a positive real number.

c) There exist real numbers x and y such that x^2 exceeds y but x is less than y . This is true, since we can take $x = 2$ and $y = 3$, for instance.

d) This says that for every pair of real numbers x and y , there exists a real number z that is their sum. In other words, the real numbers are closed under the operation of addition, another true fact. (Some authors would include the uniqueness of z as part of the meaning of the word *closed*.)

Nested Quantifiers, Questions

Negating Nested Quantifiers

1.4.30 Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

- a. $\neg \exists x \exists y p(x, y)$
- b. $\neg \forall x \exists y p(x, y)$
- c. $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
- d. $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
- e. $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$

a	$\forall x \forall y \neg p(x, y)$
b	$\exists x \forall y \neg p(x, y)$
c	$\forall y (\neg Q(y) \vee \exists x R(x, y))$
d	$\forall y (\forall x \neg R(x, y) \wedge \exists x \neg S(x, y))$
e	$\forall y (\exists x \forall z \neg T(x, y, z) \wedge \forall x \exists z \neg U(x, y, z))$

Rules of Inference

Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an argument, we mean a sequence of statements that end with a conclusion. By valid, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument..

A rule of inference, inference rule or transformation rule is a logical form consisting of a function which takes premises, analyzes their syntax, and returns a conclusion (or conclusions).

Valid Arguments in Propositional Logic

“If you have a current password, then you can log onto the network.”

“You have a current password.”

Therefore,

“You can log onto the network.”

Rules of Inference for Propositional Logic

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Rules of Inference for Propositional Logic

Modus Ponens:

The Modus Ponens rule is one of the most important rules of inference, and it states that if P and $P \rightarrow Q$ is true, then we can infer that Q will be true. It can be represented as:

Example:

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I am sleepy" $\implies P$

P	Q	$P \rightarrow Q$
0	0	0
0	1	1
1	0	0
1	1	1



Conclusion: "I go to bed." $\implies Q$.

Hence, we can say that, if $P \rightarrow Q$ is true and P is true then Q will be true.

Modus Tollens:

The Modus Tollens rule states that if $P \rightarrow Q$ is true and $\neg Q$ is true, then $\neg P$ will also be true. It can be represented as:

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I do not go to the bed." $\implies \neg Q$

Statement-3: Which infers that "I am not sleepy" $\implies \neg P$

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	0
1	1	0	0	1



Rules of Inference for Propositional Logic

Hypothetical Syllogism:

The Hypothetical Syllogism rule states that if $P \rightarrow Q$ is true whenever $P \rightarrow Q$ is true, and $Q \rightarrow R$ is true. It can be represented as the following notation:

Example:

Statement-1: If you have my home key then you can unlock my home. $P \rightarrow Q$

Statement-2: If you can unlock my home then you can take my money. $Q \rightarrow R$

Conclusion: If you have my home key then you can take my money. $P \rightarrow R$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$P \rightarrow R$	
0	0	0	1	1	1	←
0	0	1	1	1	1	←
0	1	0	1	0	1	
0	1	1	1	1	1	←
1	0	0	0	1	1	
1	0	1	0	1	1	
1	1	0	1	0	0	
1	1	1	1	1	1	←

Rules of Inference for Propositional Logic

Disjunctive Syllogism:

The Disjunctive syllogism rule state that if $P \vee Q$ is true, and $\neg P$ is true, then Q will be true. It can be represented as:

Example:

Statement-1: Today is Sunday or Monday. $\implies P \vee Q$

Statement-2: Today is not Sunday. $\implies \neg P$

Conclusion: Today is Monday. $\implies Q$

P	Q	$\neg P$	$P \vee Q$
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1

Addition:

The Addition rule is one the common inference rule, and it states that If P is true, then $P \vee Q$ will be true.

Example:

Statement: I have a vanilla ice-cream. $\implies P$

Statement-2: I have Chocolate ice-cream.

Conclusion: I have vanilla or chocolate ice-cream. $\implies (P \vee Q)$

P	Q	$P \vee Q$
0	0	0
1	0	1
0	1	1
1	1	1

Rules of Inference for Propositional Logic

Simplification:

The simplification rule states that if $P \wedge Q$ is true, then Q or P will also be true. It can be represented as:

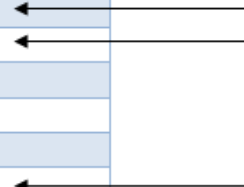
P	Q	$P \wedge Q$
0	0	0
1	0	0
0	1	0
1	1	1



Resolution:

The Resolution rule states that if $P \vee Q$ and $\neg P \wedge R$ are true, then $Q \vee R$ will also be true. It can be represented as

P	$\neg P$	Q	R	$P \vee Q$	$\neg P \wedge R$	$Q \vee R$
0	1	0	0	0	0	0
0	1	0	1	0	0	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	0	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1



Using Rules of Inference to Build Arguments

Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

Let p be the proposition “It is sunny this afternoon,” q the proposition “It is colder than yesterday,” r the proposition “We will go swimming,” s the proposition “We will take a canoe trip,” and t the proposition “We will be home by sunset.” Then the premises become $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s$, and $s \rightarrow t$. The conclusion is simply t . We need to give a valid argument with premises $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s$, and $s \rightarrow t$ and conclusion t .

We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

TABLE 1 Rules of Inference.

Rule of Inference	Tautology	Name
$\frac{p \quad p \rightarrow q}{\therefore q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Using Rules of Inference to Build Arguments

Show that the premises “If you send me an e-mail message, then I will finish writing the program,” “If you do not send me an e-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed” lead to the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed.”

Solution: Let p be the proposition “You send me an e-mail message,” q the proposition “I will finish writing the program,” r the proposition “I will go to sleep early,” and s the proposition “I will wake up feeling refreshed.” Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$. The desired conclusion is $\neg q \rightarrow s$. We need to give a valid argument with premises $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$ and conclusion $\neg q \rightarrow s$.

This argument form shows that the premises lead to the desired conclusion.

Step	Reason
1. $p \rightarrow q$	Premise
2. $\neg q \rightarrow \neg p$	Contrapositive of (1)
3. $\neg p \rightarrow r$	Premise
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Premise
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)

TABLE 1 Rules of Inference.		
Rule of Inference	Tautology	Name
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Resolution

Computer programs have been developed to automate the task of reasoning and proving theorems. Many of these programs make use of a rule of inference known as resolution. This rule of inference is based on the tautology Links.

The Resolution Principle are used to prove whether an argument is CORRECT OR NOT.

Literal: A variable or negation of a variable is called a literal.

Clause: A clause is a disjunction of literals.

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r).$$

The final disjunction in the resolution rule, $q \vee r$, is called the **Resolvent**.

When we let $q = r$ in this tautology, we obtain $(p \vee q) \wedge (\neg p \vee q) \rightarrow q$.

Furthermore, when we let $r = F$, we obtain $(p \vee q) \wedge (\neg p) \rightarrow q$ (because $q \vee F \equiv q$), which is the tautology on which the rule of disjunctive syllogism is based.

Resolution

Use resolution to show that the hypotheses “Jasmine is skiing or it is not snowing” and “It is snowing or Bart is playing hockey” imply that “Jasmine is skiing or Bart is playing hockey.”

Solution: Let p be the proposition “It is snowing,” q the proposition “Jasmine is skiing,” and r the proposition “Bart is playing hockey.” We can represent the hypotheses as $\neg p \vee q$ and $p \vee r$, respectively. Using resolution, the proposition $q \vee r$, “Jasmine is skiing or Bart is playing hockey,” follows.

Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

Solution: We can rewrite the premises $(p \wedge q) \vee r$ as two clauses, $p \vee r$ and $q \vee r$. We can also replace $r \rightarrow s$ by the equivalent clause $\neg r \vee s$. Using the two clauses $p \vee r$ and $\neg r \vee s$, we can use resolution to conclude $p \vee s$.

Fallacies

There are some common arguments which are invalid. Though at first glance, they seem right. Such a type of argument is known as Fallacy.

The proposition $((p \rightarrow q) \wedge q) \rightarrow p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology.

In other words, they treat the argument with premises $p \rightarrow q$ and q and conclusion p as a valid argument form, which it is not. This type of incorrect reasoning is called **the fallacy of affirming the conclusion**.

Fallacies

Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics.

You learned discrete mathematics.

Therefore,

you did every problem in this book.

Solution: Let p be the proposition “You did every problem in this book.” Let q be the proposition “You learned discrete mathematics.” Then this argument is of the form: if $p \rightarrow q$ and q , then p .

This is an example of an incorrect argument using the fallacy of affirming the conclusion.

Indeed, it is possible for you to learn discrete mathematics in some way other than by doing every problem in this book. (You may learn discrete mathematics by reading, listening to lectures, doing some, but not all, the problems in this book, and so on.)

The proposition $((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference.

This type of incorrect reasoning is called the fallacy of denying the hypothesis.